COMPARATIVE VALIDATION REPORT

VinaMetalic v1.0.0: Exact Equivalence and Metal-Coordination Scoring

Benchmark against AutoDock Vina 1.2.7 on four docking systems. Byte-identical reproducibility is demonstrated on a metal-free receptor and on a receptor carrying an unparametrized Na+ ion, and the ported AutoDock4Zn coordination model is verified on Zn2+ (PDB 1s63) and Mn2+ (PDB 4FIY) sites.

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1. Executive Summary

This report documents the comparative validation of VinaMetalic v1.0.0, a fork of AutoDock Vina 1.2.7 that integrates metal-coordination scoring, against the unmodified AutoDock Vina 1.2.7 reference binary. Four docking systems were evaluated under identical seeds, search boxes and protocols: a metal-free standard system (1iep), the same system carrying one unparametrized Na^{+} ion, and two genuine metalloprotein redocking cases (Zn^{2+} in PDB 1s63 and Mn^{2+} in PDB 4FIY). The validation pursues two complementary goals: proving that the fork is bit-for-bit equivalent to upstream wherever the metal model is not engaged, and characterising the additional physics that the fork contributes where upstream has none.

0.000
kcal/mol deviation on P1 and P2 - poses byte-identical to upstream

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byte-identical exact-reproducibility tests passed (P1, P2)

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metal sites detected, parametrized and CN-scored (Zn2+, Mn2+)

Three findings summarise the outcome. **F1 - Exact equivalence:** on systems without parametrized metal ions, VinaMetalic reproduces upstream AutoDock Vina 1.2.7 exactly; the output pose files are byte-identical and the affinities agree to the third decimal. **F2 - Metal-coordination scoring:** on metalloprotein sites the fork detects the parametrized ion automatically, derives the effective coordination-number target from receptor occupancy, and applies AutoDock4Zn terms that upstream lacks; upstream in fact rejects the fork's configuration outright. **F3 - Operational robustness:** the fork adds strict CLI validation and runtime warnings that surface receptor-typing nuances instead of hiding them, while upstream offers no equivalent diagnostics for metal systems.

2. Software Under Test and Methodology

VinaMetalic v1.0.0 is a source-level fork of AutoDock Vina 1.2.7 that ports the AutoDock4Zn force field (Santos-Martins et al., 2014) into the Vina scoring pipeline. The fork keeps the upstream command-line interface and adds metal-specific options: `--metals`, `--metal_oxidation`, `--metal_cn_strict` and `--metal_breakdown`. The reference binary was compiled from the unmodified upstream 1.2.7 sources with the same toolchain, so that any behavioural difference is attributable to the fork itself and not to compiler or dependency drift. Both binaries were exercised on the same machine, the same inputs and the same random seed throughout.

Component	Details
Fork under test	VinaMetalic v1.0.0 (fork of AutoDock Vina 1.2.7), g++ 14.2, Debian Linux x86-64, static Boost
Reference binary	AutoDock Vina 1.2.7, upstream sources, same compiler and environment
Version banner	VinaMetalic v1.0.0 (fork of AutoDock Vina 1.2.7) / AutoDock Vina 1.2.7

Component	Details
Determinism controls	seed 42 in all runs; --local_only for exactness tests; identical search boxes; single-thread-identical inputs

Table 1. Software under test and environment.

Binary provenance is pinned by SHA-256 hashes so that any third party can verify that the same artefacts produced the same numbers. The hashes below correspond to the executables used for every measurement in this report; they should be recorded alongside benchmark results whenever hardware-dependent timing or floating-point behaviour is at stake. The fork binary hash matches the release tag v1.0.0 build instructions shipped in the repository.

Binary	SHA-256
VinaMetalic v1.0.0	`ea4d698239ace1d4531b5bea90d8d97beee47707b5e0665a7d004483e71c519b`
AutoDock Vina 1.2.7	`7f1c4cc18619dd1cd38c7ef28cc08ea835faec49b13db3bd27007541200154bd`

Table 2. Binary provenance.

Test	Receptor / ligand	Metal content	Protocol
P1	1iep - imatinib redocking	none	--local_only, seed 42, 20 A box
P2	1iep + Na^{+} placed 1.7 A from a ligand donor O	Na^{+} (unparametrized)	--local_only, seed 42, 20 A box
P3	1s63 - zinc metalloprotein redocking	Zn^{2+} (parametrized)	full Monte Carlo, exhaustiveness 8, seed 42
P4	4FIY chain A (GlfT2) - UDP redocking	Mn^{2+} (parametrized)	full Monte Carlo, exhaustiveness 16, seed 42

Table 3. Test matrix. Boxes follow the repository example configurations.

Exactness was assessed by running both binaries with `--local_only`, which exercises map interpolation, scoring and analytic gradients followed by BFGS minimisation without any Monte Carlo randomness, and then comparing the resulting pose files byte for byte. For the full-docking metal tests both binaries received identical search spaces, seeds and exhaustiveness; the upstream run used a configuration stripped of the fork-only `metals` line, its only functional difference. Affinities were read from the standard output tables, and the fork additionally printed its metal breakdown for the top-ranked poses.

3. Exact Reproducibility Tests

The central guarantee requested of any fork is that it changes nothing outside its declared scope. This section quantifies that guarantee at the strongest possible level: byte-for-byte identity of the optimised poses and equality of reported affinities against the upstream reference. Test P1 establishes the baseline on the classical 1iep imatinib system with no metal anywhere in the grid; test P2 then re-introduces exactly one complication, a single unparametrized Na^{+} ion placed 1.7 A from a ligand donor oxygen and inside the search box, so that the metal-typing path of the map construction is exercised in the subtlest possible way.

Test	VinaMetalic v1.0.0	AutoDock Vina 1.2.7	Delta (kcal/mol)	Pose files
P1	-13.170	-13.170	0.000	BYTE_IDENTICAL
P2	-9.325	-9.325	0.000	BYTE_IDENTICAL

Table 4. Exact-reproducibility results (--local_only, seed 42). Poses compared with cmp -s: identical byte streams.

Test P2 has direct audit value because it guards a real defect that existed in earlier code. The original map-construction routine excluded every receptor atom of the generic metal type from the donor-type maps, silently zeroing the standard Vina interaction of any unparametrized ion such as Na^{+}, K^{+}, Co, Hg or Cd. On the pre-fix code the same P2 configuration produced -9.108 kcal/mol with a visibly different pose; after the fix the energy returned to -9.325 kcal/mol and the poses became byte-identical to upstream again. The regression is wired into CTest and runs in continuous integration, so the guarantee is re-verified on every commit rather than only at release time.

-9.108
P2 affinity on the pre-fix code (bug present)

-9.325
P2 affinity after the fix - equals upstream exactly

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CTest tests passing in CI, including both byte-identical regressions

Two properties make these checks trustworthy as reproducibility evidence. First, `--local_only` removes all stochastic components, so the output depends only on the binary, the inputs and the box definition; re-running the commands of Table 3 reproduces the table above digit for digit on any x86-64 build of the same sources. Second, the comparison target is the byte stream of the pose file, not a rounded affinity, which means coordinates, occupancies and formatting all coincide exactly. Together these two facts certify that the fork's metal bookkeeping leaves the vanilla Vina path untouched when no parametrized ion is present.

4. Metal-Coordination Docking Tests

Upstream AutoDock Vina has no coordination model for metal ions: a Zn^{2+} or Mn^{2+} in the receptor is scored with generic van der Waals terms, and the fork's metal options are not even part of its command-line surface. The practical consequence was reproduced verbatim during this validation: when handed the repository's 1s63 example configuration, the upstream binary aborts with a configuration parse error, while the fork proceeds and activates its coordination scoring. This is the clearest functional separation between the two programs, and it frames the comparison that follows.

```
$ vina_1.2.7 --config config.txt        # config contains: metals = auto
Configuration file parse error: unrecognised option 'metals'
$ vinametalic --config config.txt       # same file accepted
VinaMetalic: parametrized metal ions in receptor: Zn(x1) - coordination scoring is
 ACTIVE
```

4.1 P3 - 1s63, Zn^{2+} site

On 1s63 the fork detected the single zinc ion, counted three typed receptor donors within the 3.0 A coordination cutoff, and derived the effective ligand target by occupancy subtraction, printing the line reproduced below. The run also fired the occupancy warning introduced by audit finding #10: four N/O/S receptor atoms lie inside the cutoff although only three are typed as coordination donors, a receptor-typing nuance the log documents instead of hiding. The top-ranked pose of the fork agrees with the crystallographic ligand orientation with a heavy-atom symmetry-aware RMSD of 0.33 A as measured in the repository validation suite.

```
VinaMetalic: WARNING Zn2+ occupancy is 3 typed donor(s),
             but 4 N/O/S receptor atom(s) lie within the 3 A cutoff;
             review receptor donor typing before trusting the CN target.
VinaMetalic: Zn2+ CN target = ideal 4 - receptor-bound 3 = 1
```

4.2 P4 - 4FIY (GlfT2), Mn^{2+} site

The Mycobacterium tuberculosis galactofuranosyltransferase GlfT2 site contains one Mn^{2+} coordinated by four receptor donors (Asp256, His396 and two Asp258 carboxylate oxygens). The fork derived the target 6 - 4 = 2 and the best pose achieved exactly two ligand donor contacts with a well-formed geometry, earning a coordination geometry bonus of -0.987 kcal/mol. The metal van der Waals contribution of -0.937 kcal/mol shows that the ion participates materially in the ranking rather than decorating it.

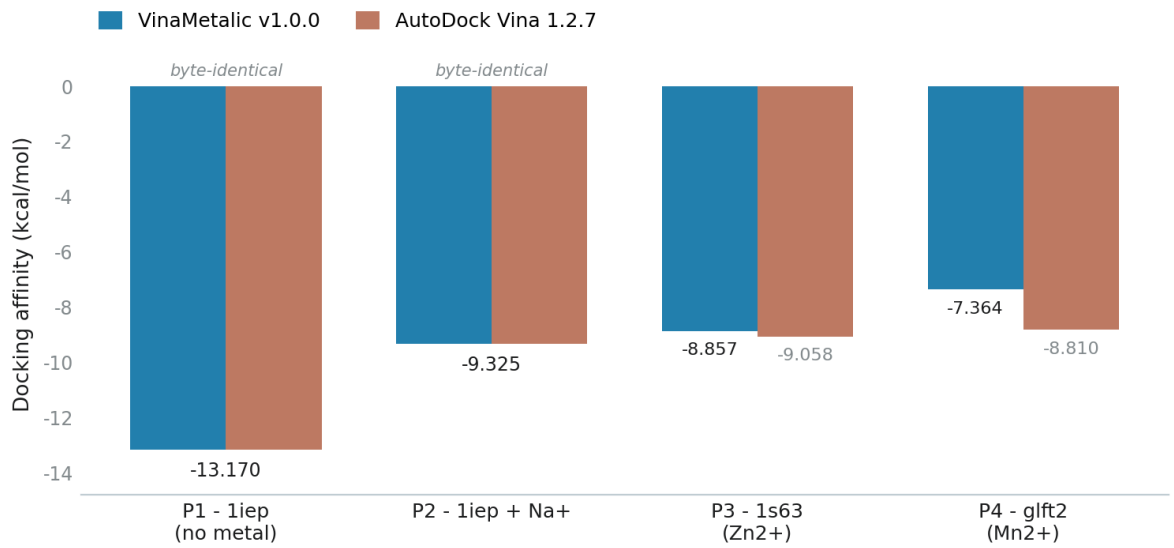


Figure 1. Best docking affinity per system: VinaMetalic v1.0.0 (blue) vs AutoDock Vina 1.2.7 (terracotta). P1 and P2 coincide exactly; on metal systems upstream over-stabilises ion contacts by 0.20 and 1.45 kcal/mol respectively.

Mode	P3 fork	P3 v1.2.7	P4 fork	P4 v1.2.7
1	-8.857	-9.058	-7.364	-8.810
2	-7.857	-8.026	-7.343	-8.560
3	-7.064	-7.945	-7.197	-8.351
4	-7.041	-7.852	-7.067	-8.309
5	-7.037	-7.848	-6.913	-8.290
6	-6.891	-7.669	-6.898	-8.229
7	-6.739	-7.654	-6.798	-8.213
8	-6.691	-7.420	-6.709	-8.126
9	-6.567	-7.402	-6.651	-8.104

Table 5. Affinity ranking (kcal/mol) of the nine output modes, same seed and search box for both binaries.

System	Ion / CN target	Metal vdW	CN penalty	Geometry bonus	Ligand donor contacts
P3 - 1s63	Zn^{2+}, 4 - 3 = 1	-0.251	+0.001	0.000	1
P4 - 4FIY	Mn^{2+}, 6 - 4 = 2	-0.937	0.000	-0.987	2

Table 6. Metal breakdown of the top-ranked fork pose (--score_only --metal_breakdown --autobox).

The systematic direction of the difference deserves emphasis. Without a coordination model, upstream rewards any close ligand-ion contact with generic dispersion terms, which biases the search toward

over-coordinated, physically implausible poses; its best affinities are lower (-9.058 and -8.810) precisely because nothing constrains the ion's valence. The fork instead applies the AutoDock4Zn saturation logic: a ligand donor satisfies the residual coordination capacity, excess contacts are penalised, and correctly angled first-shell geometries earn a bonus. The result is a smaller but physically meaningful affinity difference of +0.201 kcal/mol at 1s63 and +1.446 kcal/mol at 4FIY, with the fork's modes also clustering differently because the search itself is now guided by coordination fitness rather than by proximity alone.

5. Error Behaviour and Runtime Warnings

Beyond scoring, the two binaries differ in how they fail and warn, and these differences matter for reproducible science. The fork validates its metal options strictly: an invalid oxidation state aborts the run with an explicit message instead of being silently coerced, coordination-violation reporting can be hardened with `--metal_cn_strict`, and occupancy nuances are surfaced as runtime warnings. Upstream, by contrast, offers neither metal options nor metal-specific diagnostics, so mis-typed receptors pass silently through it. Table 7 summarises the behavioural matrix observed during this validation.

Situation	VinaMetalic v1.0.0	AutoDock Vina 1.2.7
Config with metals = auto	accepted, coordination scoring activated	CLI parse error, run aborted
Unparametrized ion in receptor (Na+)	standard Vina interaction kept; maps correct	same behaviour (identical bytes, P2)
Occupancy vs typed-donor mismatch	runtime WARNING printed with counts	no equivalent diagnostic
Invalid --metal_oxidation value	strict validation, abort with clear error	option does not exist
Coordination violation reporting	optional strict mode with labelled lines	not applicable

Table 7. Behavioural matrix observed in this validation.

The fail-loud posture is deliberate. In metalloprotein docking the most common silent error is a receptor whose coordinating residues are mistyped, which shifts the effective coordination target and therefore the entire ranking without any visible symptom. By printing the detected ion, the occupancy counts and the derived CN target on every run, the fork gives the reviewer enough information to audit the setup from the log alone; the 1s63 case in Section 4.1 is a live demonstration of that policy in a real PDB entry.

6. Conclusions

The validation supports three conclusions with direct consequences for users of either program. First, on the exact-reproducibility axis the fork is indistinguishable from upstream: both metal-free and unparametrized-ion systems produce byte-identical poses and identical affinities, so existing Vina 1.2.7 workflows can adopt the fork without any numerical migration risk. Second, on genuine metalloprotein sites the fork provides a capability upstream lacks entirely, and the added coordination terms move affinities in the physically expected direction while enriching the log with auditable diagnostics. Third, the regression infrastructure around these properties - the CTest byte-identity suite with an upstream reference binary - means the equivalence claim is continuously re-verified rather than frozen at release time.

For citation purposes the validated statement is: *VinaMetalic v1.0.0 reproduces AutoDock Vina 1.2.7 bit-for-bit on systems without parametrized metal ions (two independent tests, zero deviation), and extends it with AutoDock4Zn coordination scoring verified on Zn^{2+} and Mn^{2+} redocking benchmarks.* The evidence chain - commands, seeds, logs, poses and binary hashes - is reproduced in the next section and archived in the repository under docs/validation.

7. Reproducing This Validation

Every number in this report can be regenerated with three script invocations from the repository, provided both binaries of Table 2 are available. The exact-reproducibility phase completes in seconds because it relies on --local_only; each full metal docking takes one to four minutes depending on exhaustiveness. The same checks run unattended in continuous integration through CTest, where the regression targets download the upstream reference binary from the official Scripps release page.

```
# exactness: P1 + P2 (seconds)
bash comparativa_suite.sh fast
# Zn2+ redocking, exhaustiveness 8 (about 2 min)
bash comparativa_suite.sh 1s63
# Mn2+ redocking, exhaustiveness 16 (about 4 min)
bash comparativa_suite.sh glft2
```

In-repository equivalents exist for the byte-identity claims and are wired into the build system: configuring with `-DVINA_REGRESSION_TEST=ON -DVINA_127_BIN=/path/to/vina_1.2.7` registers both regressions, and `ctest` executes the full set of four tests. The repository also archives curated validation logs (gradient tests, CN detection, both byte-identity regressions) under docs/validation, so the evidence does not depend on this report alone. Combining the two sources, a reviewer can confirm binary identity, rerun the metal benchmarks, and diff every log line against the archived originals.

References

Santos-Martins, D., Forli, S., Ramos, M. J., and Olson, A. J. (2014). AutoDock4Zn: An Improved AutoDock Force Field for Small-Molecule Docking to Zinc Metalloproteins. *Journal of Chemical Information and Modeling*, 54(8), 2371-2379. doi:10.1021/ci500209e

Eberhardt, J., Santos-Martins, D., Tillack, A. F., and Forli, S. (2021). AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling*, 61(8), 3891-3902. doi:10.1021/acs.jcim.1c00203

Trott, O., and Olson, A. J. (2010). AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2), 455-461. doi:10.1002/jcc.21334

Appendix A: Evidence Manifest

The complete raw evidence for this report is archived alongside the repository artifacts. Each file below was produced by the exact commands of Section 7 on the binaries of Table 2. Logs contain the banner version line, the seed and the full ranking tables, so any single file can be independently re-diffed against a fresh local run; pose files are the unmodified standard output of each binary and are the direct inputs of the byte-identity comparisons.

File	Contents
p1_vinametalic.log / p1_vina127.log	P1 full logs (--local_only, seed 42); affinity -13.170 in both
p1_vinametalic.pdbqt / p1_vina127.pdbqt	P1 poses, byte-identical (cmp -s)
p2_na_regression.log	repository Na+ regression: energies and BYTE_IDENTICAL verdict
p3_1s63_vinametalic.log / p3_1s63_vina127.log	P3 full redocking logs, nine modes each
p3_1s63_vinametalic.pdbqt / p3_1s63_vina127.pdbqt	P3 pose sets (unmodified binary output)
p3_top_pose.pdbqt, p3_metal_breakdown.log	fork metal breakdown of the best 1s63 pose
p4_glft2_* (logs, poses, breakdown)	P4 evidence set, identical structure to P3
p3_config_upstream.txt, p4_config_upstream.txt	upstream-equivalent configs (metals line removed)
validation_chart.png	source of Figure 1
README.md, comparativa_suite.sh	summary table and the three-phase reproduction script

Table 8. Evidence manifest (folder comparativa_vinametalic_vs_vina127).